

# Closure under an Invariant Form: one principle across geometry, arithmetic, and physics

(VFD programme — working draft)

2026

**Read first (status).** This paper states a **lens**, not a theorem, and a **map**, not a manifesto. It identifies one mathematical pattern — *a structure closes when it is positive under its proper invariant form* — and grades three instances honestly: **geometry** ( $H_4 \rightarrow E_8$ ) where the form *delivers* closure (exact, proved); **arithmetic** (RH) where the form is known and finite-positive but full closure is RH (open); and **physics** (boundary scattering) where the architecture is shared but *does not transfer* (analogy). **The right form is necessary, not sufficient: RH is the standing counterexample to “form  $\Rightarrow$  closure”.** There is **no cross-domain implication**: the instances share the principle, not a proof. No proof of RH, and no claim that any physical model implies or is implied by arithmetic, is made.

## Abstract

We extract one structure from three otherwise unrelated computations of the VFD programme: in each, a *naive* bilinear form fails (fractional residual, non-positive kernel, cosmetic geometry) and a *completed invariant form* repairs it. We formalise this as a single object — a *closure datum*  $(V, B, C)$  — and a single principle: a structure exists/is stable iff it is positive (closes) under its proper form  $B$ . We prove the enabling fact (self-adjointness and positivity are *form-relative*) and demonstrate it exactly on multiplication by  $\varphi$ . We then place three instances on the map: the  $H_4 \rightarrow E_8$  folding (the form delivers closure, exactly), the icosian witness operator for the Riemann Hypothesis (the form is the explicit formula; full positivity is RH, open), and boundary scattering (a shared determinant/flux architecture, analogy only). The unification is the *principle with its instances*, one of which is fully proved; it is explicitly *not* a chain of cross-domain implications. We make no claim about the Riemann Hypothesis.

## 1 The closure datum and the principle

**Definition 1** (closure datum). A closure datum is a triple  $(V, B, C)$ : a module or Hilbert space  $V$ , an invariant symmetric/Hermitian form  $B$  on  $V$ , and a closure operator  $C$  (a structure map whose fixed points are the “closed” configurations). A configuration  $X$  closes when it is  $B$ -positive (equivalently a fixed point of  $C$ ).

**Principle 2** (closure under an invariant form). Across the instances below,  $X$  exists / is stable  $\iff X$  is  $B$ -positive (closes). The principle is shared; the form  $B$  is domain-specific. **Possessing the canonical form is necessary, not sufficient**: completing to  $B$  can convert apparent leakage into closed structure, but does not force closure (§4 is the counterexample).

The content of the paper is (i) the enabling lemma (§2); (ii) three graded instances (§§3–5); (iii) the precise statement of what relates them and what does not (§6).

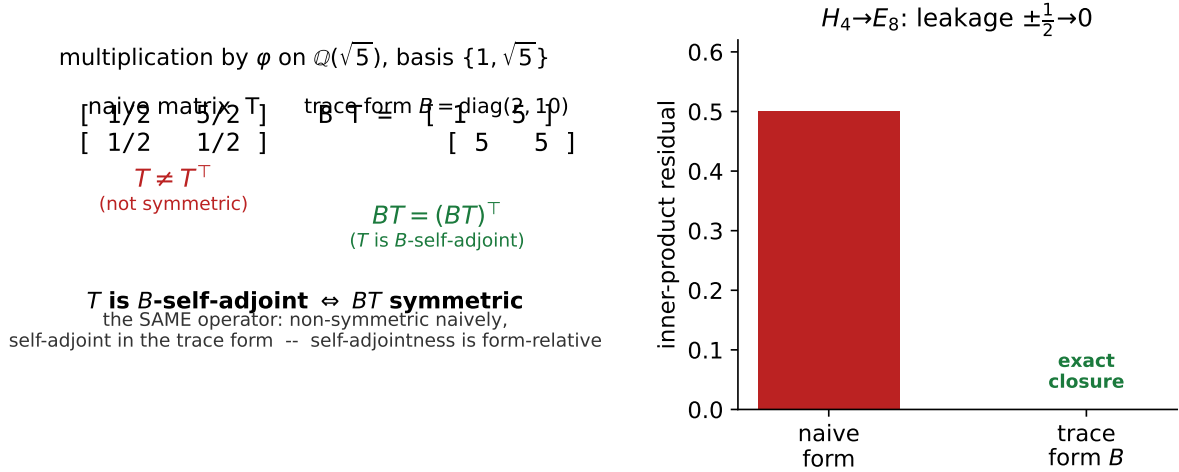


Figure 1: Form-relativity (exact). The multiplication-by- $\varphi$  operator is not symmetric under the naive form but is self-adjoint under the trace form  $B$ ; equivalently the  $H_4 \rightarrow E_8$  inner-product residual closes from  $\pm \frac{1}{2}$  to 0.

## 2 Self-adjointness is form-relative

**Lemma 3** (form-relativity). *Let  $B$  be an invertible symmetric form and  $T$  a linear operator. The  $B$ -adjoint is  $T^{\dagger B} = B^{-1}T^\top B$ , and  $T$  is  $B$ -self-adjoint iff  $BT$  is symmetric.*

*Proof.*  $\langle Tx, y \rangle_B = (Tx)^\top B y = x^\top T^\top B y$  and  $\langle x, Sy \rangle_B = x^\top B S y$ ; equality for all  $x, y$  gives  $T^\top B = BS$ , i.e.  $S = B^{-1}T^\top B$ . Then  $T^{\dagger B} = T$  iff  $T^\top B = BT$  iff  $(BT)^\top = B^\top T = BT$  (as  $B$  is symmetric), i.e.  $BT$  symmetric.  $\square$

**Remark 4** (exact anchor). Multiplication by  $\varphi = \frac{1+\sqrt{5}}{2}$  on  $\mathbb{Q}(\sqrt{5})$  in the basis  $\{1, \sqrt{5}\}$  has matrix  $T = \begin{bmatrix} 1/2 & 5/2 \\ 1/2 & 1/2 \end{bmatrix}$ , which is *not* symmetric. Under the trace form  $B(x, y) = \text{Tr}(xy) = \text{diag}(2, 10)$  we get  $BT = \begin{bmatrix} 1 & 5 \\ 5 & 5 \end{bmatrix}$ , symmetric: by Lemma 3,  $\varphi$  is  $B$ -self-adjoint. The *same* operator is non-symmetric naively and self-adjoint in the trace form — self-adjointness is form-relative, demonstrated exactly (Fig. 1).

## 3 Instance G — geometry: the form delivers closure (exact)

The icosian/ $H_4$  data embeds into  $E_8$  only after the naive map  $x \mapsto (x, \sigma x)$  (inner products  $\pm \frac{1}{2}$ , not even-integral) is completed to the *golden trace form*  $B(x, y) = \text{Tr}_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(c \langle x, y \rangle)$  with  $c = 1 + 1/\sqrt{5}$  (so  $\text{Tr} c = 2$ ), *derived* by the norm-2 and integrality constraints. Under  $B$ ,

$\text{shell}_1 \cup (1/\varphi) \text{shell}_1 = 240 \text{ roots}$ ,  $\|\cdot\|^2 = 2$ , root graph  $(56, 56, 126, 1, 1)$ , rank 8, even integral,

i.e. the canonical  $E_8$  (Fig. 2). The residual closes  $\pm \frac{1}{2} \rightarrow 0$ : *the form delivers closure, exactly*. This is the known Wilson / Moody–Patera construction [1, 2], verified; we use it as the one instance where Principle 2 is *realised*, not merely posited.

$H_4 \rightarrow E_8$  (EXACT): 240 roots in the Coxeter plane  
 $\text{shell}_1 \cup (1/\varphi)\text{shell}_1$ ,  $\text{norm}^2 = 2$ , rank 8, even integral

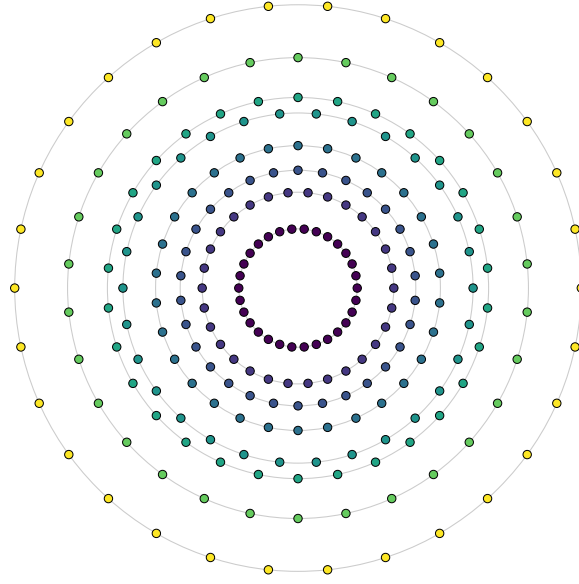


Figure 2: Instance G (EXACT). The 240  $E_8$  roots,  $\text{shell}_1 \cup (1/\varphi)\text{shell}_1$ , in the Coxeter plane: eight concentric rings of 30 (verified  $240 = 8 \times 30$ ). Under the golden trace form the structure is even-integral  $E_8$  — the form closes it.

#### 4 Instance A — arithmetic: the form is known, closure is RH (open)

The same pattern, on the scale axis. For the self-dual cuspidal  $L$ -function  $L(s, \pi)$  of the icosian Brandt face at level  $\mathfrak{p}_{31}$  over  $K = \mathbb{Q}(\sqrt{5})$ , the naive prime-only kernel is non-positive and divergent; completing to the Weil/Connes form — archimedean boundary  $A_\infty$  (gamma factor, from Tate’s local theory on the scale axis) minus the geometric prime residue  $A_P$  (scale-translations weighted by the icosian Brandt eigenvalues  $a_q$ ) — yields a self-adjoint operator

$$A = A_\infty - A_P, \quad \langle h, Ah \rangle = \sum_{\rho} |\widehat{h}(\gamma_\rho)|^2, \quad \boxed{A \geq 0 \iff \text{RH}(L)}$$

(the identification lemma and the equivalence are established in the companion [3]; Fig. 3). Here the canonical form is the explicit formula itself, and it is *finite-positive* on every tested band — yet **full positivity is exactly RH(L), and is open**. This is the standing counterexample to a guarantee: the right form is present and finite-positive, and closure is still not delivered. (Scope:  $\text{RH}(L)$  is GRH for this one cuspidal  $L$ , *not* classical  $\zeta$ ; no RH proof is claimed.)

#### 5 Instance P — physics: shared architecture, no transfer (analogy)

A third appearance is dynamical. A static “funnel” geometry is cosmetic (the radius is a bijection, no invariant); completing to the *scattering determinant* / Jost function with a greybody ( $\Gamma$ -ratio)

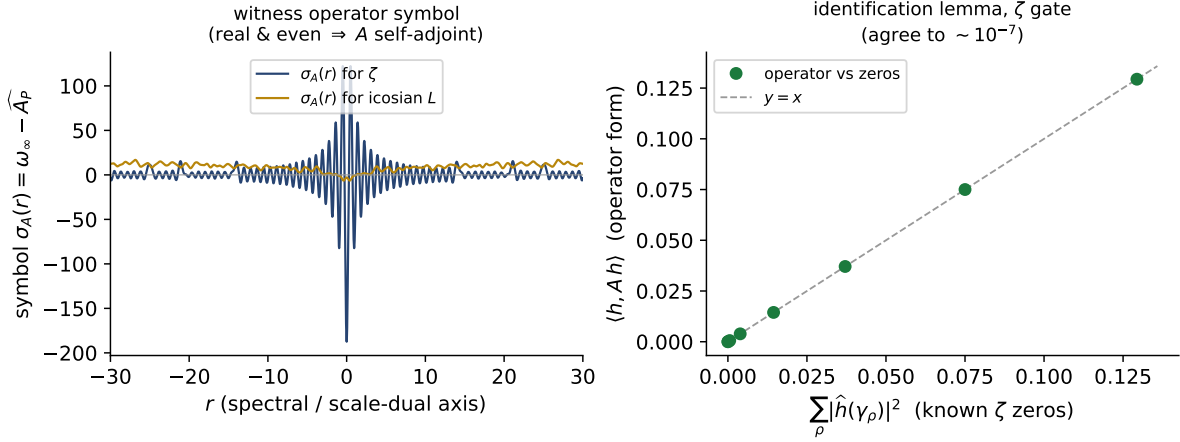


Figure 3: Instance A (OPEN). Left: the witness-operator symbol  $\sigma_A(r) = \omega_\infty - \widehat{A}_P$  for  $\zeta$  and for the icosian  $L$  — real and even, so  $A$  is self-adjoint. Right: the operator form reproduces the sum over the known Riemann zeros to  $\sim 10^{-7}$  (the identification lemma). Positivity of  $A$  for *all*  $h$  is the wall = RH( $L$ ); it is not shown.

factor produces a genuine architecture — determinant  $\rightarrow$  log-derivative (digamma)  $\rightarrow$  resonances on lines  $\rightarrow$  stability — the known trace-formula / Lax–Phillips picture. For a *real* (self-adjoint) potential this gives damped quasinormal modes; but the arithmetic instance of the *same* statement is Hilbert–Pólya, i.e. RH. So the form/architecture is *parallel*, and the closure does *not* transfer: the toy is provable because its potential is self-adjoint; the arithmetic self-adjointness is open. This instance is **analogy only**. (The  $\sigma$ -paired closure form also appears in a separate cosmological *model* [4]; that is a physical model on the same geometry, firewalled from the arithmetic — see §6.)

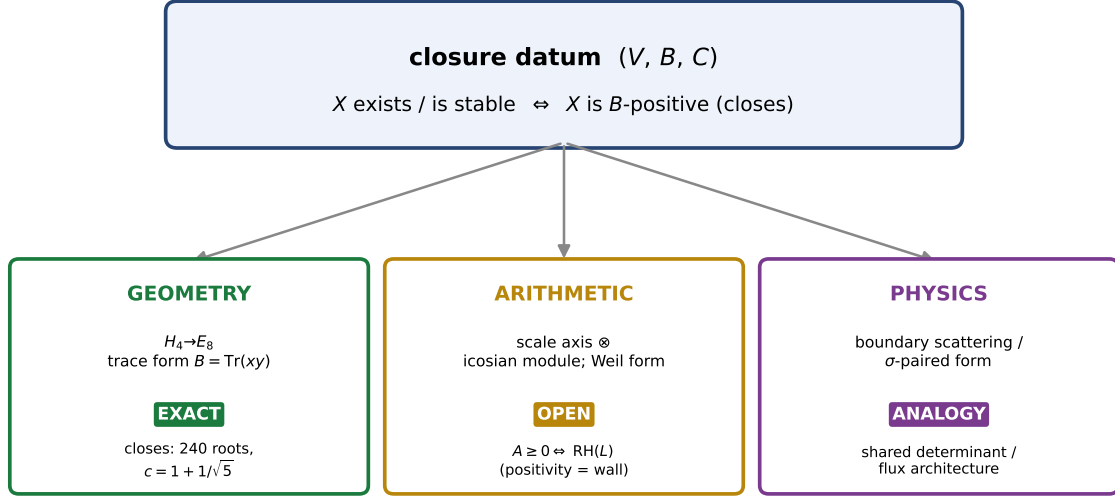
## 6 The unification: one principle, three forms, no implication

Putting the instances on one map (Fig. 4):

| instance   | closure datum $(V, B)$                         | “closes” means                          | status                 |
|------------|------------------------------------------------|-----------------------------------------|------------------------|
| geometry   | $H_4$ data, golden trace form                  | even-integral $E_8$ (240 roots)         | EXACT (proved)         |
| arithmetic | scale axis $\otimes$ icosian module, Weil form | $A \geq 0 \iff \text{RH}(L)$            | OPEN (positivity = RH) |
| physics    | boundary geometry, scattering/ $\sigma$ form   | self-adjoint $\Rightarrow$ stable modes | ANALOGY (no transfer)  |

**What relates them.** The shared object is the closure datum and Principle 2: in each domain, *exists/stable*  $\iff B$ -positive. The relation lives at the level of the abstract datum  $(V, B, C)$ , realised independently in each domain; the recurring concrete structures ( $E_8/H_4$ ,  $\sigma$ -pairing, the closure operator) are a shared *vocabulary*.

**What does not relate them (firewall).** There is *no* implication geometry  $\Rightarrow$  arithmetic  $\Rightarrow$  physics. The geometry does not prove the arithmetic; no physical model implies or is implied by  $\zeta$ . Each instance has its own, independent status. This is deliberate and load-bearing: because no instance props up another, no instance can drag another down. The proved geometric instance shows the principle is *not vacuous*; the arithmetic instance gives it a precise equivalent of RH; the physical instance shows reach. One proved, one equivalence, one analogy.



NO cross-domain implication: instances share the *PRINCIPLE* (form-relative closure), not a proof.

right form is NECESSARY, not SUFFICIENT (RH is the standing counterexample: form known + finite-positive, closure still open)

Figure 4: The map. One closure datum; three graded instances. Geometry: the form delivers closure (exact). Arithmetic: the form is known, closure is RH (open). Physics: shared architecture, analogy only. The instances share the *principle*, not a proof: there is no cross-domain implication, and the right form is necessary, not sufficient.

## 7 The object behind the scale axis

The arithmetic instance lives on a scale axis whose two ends and equator are one compactified object  $\mathcal{V}_1 = ([-\infty, +\infty] \times S^1, \pi, Q)$ ,  $\pi(t, \theta) = e^{t/2} e^{i\theta}$ : the point ( $t \rightarrow -\infty$ ), the unit circle ( $t = 0$ , read as the critical line), and infinity ( $t \rightarrow +\infty$ ) are three boundary cuts of one object, with the mirror  $t \mapsto -t$  the functional equation and the closure law  $Q[\Phi] \geq 0$  the positivity (Fig. 5). This is the interpretive picture that motivates the arithmetic instance; it is not load-bearing for any theorem here.

## 8 Scope

We state a form-relativity lens (Lemma 3, exact), one proved instance where the form delivers closure ( $H_4 \rightarrow E_8$ ), one instance where the canonical form is known and finite-positive yet closure is RH (open), and one analogy (scattering). We do *not* prove RH; we do *not* claim the principle forces closure (RH refutes the guarantee version); we do *not* claim any cross-domain implication, nor any connection between a physical model and  $\zeta$ . The unification is the principle with its instances — a lens and a search heuristic, with one exact exemplar — not a theorem that unifies by proof.

the closure object  $\mathcal{V}_1 = ([-\infty, +\infty] \times S^1, \pi, Q)$ : one object, three boundary cuts

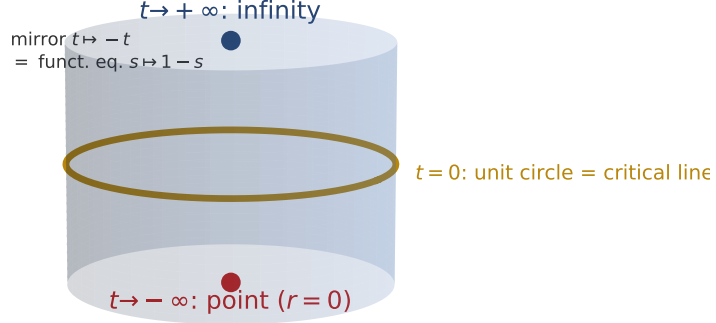


Figure 5: The closure object  $\mathcal{V}_1$  (interpretation). One compactified scale–phase object; point, equator (critical line), and infinity are its three boundary cuts; the closure law is  $Q[\Phi] \geq 0$ .

## References

- [1] R. A. Wilson, *The geometry of the Hall–Janko group as a quaternionic reflection group*, and related icosian/ $E_8$  constructions.
- [2] R. V. Moody and J. Patera, *Quasicrystals and icosians*, J. Phys. A **26** (1993).
- [3] VFD programme, *A Witness Operator on the Scale Axis* (companion draft):  $A = A_\infty - A_P$ ,  $\langle h, Ah \rangle = \sum_\rho |\widehat{h}(\gamma_\rho)|^2$ ,  $A \geq 0 \iff \text{RH}(L)$ ; geometric Brandt coefficients; numerical corroboration.
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